Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Fair test

Meaning in my words How would you explain fair test to a friend?	**How can you tell?** To test the effect of mass, what will you change and what will you keep the same?
Example or sketch How could you compare a light cart and a heavier cart fairly?	**Non-example and why** One run uses a heavier cart and a stronger pull. Why can't you tell what mass did?

Changed variable

Meaning in my words How would you explain changed variable to a friend?	**How can you tell?** When you add blocks to the cart to test the effect of mass, what did you change?
Example or sketch How could you label the runs so your partner can see exactly what varied?	**Non-example and why** The release point drifted accidentally. Is that the variable you meant to test?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Controls

Meaning in my words What are the controls in an experiment?	**How can you tell?** You test different cart masses. Why use the same hanging weight each time?
Example or sketch Name two things you would set the same before each release.	**Non-example and why** A partner lists cart mass as a control while adding mass each trial. Where does it belong?

Evidence

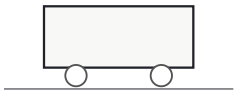
Meaning in my words How would you explain evidence to a friend?	**How can you tell?** What reading supports the claim that the cart reached 4 m in 2 s?
Example or sketch How could two measurements support a claim that one cart accelerated more?	**Non-example and why** “I expected our cart to win.” Does that tell us which cart crossed first?

Doodle the idea

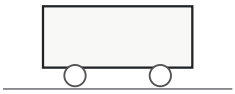
SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw two masses with the same force. Label what changes, what stays fixed, and which accelerates more.

Lower mass



Higher mass



My marks mean

Same push. Different mass.

11

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Compare velocity changes when masses differ.

CHOOSE ONE JOB TOGETHER

TEST ENGINEER ☐

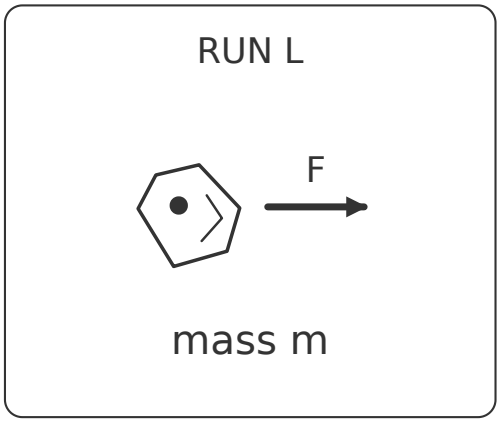
Compare the two test runs. Tell the crew which body changes velocity more and use the masses to explain why.

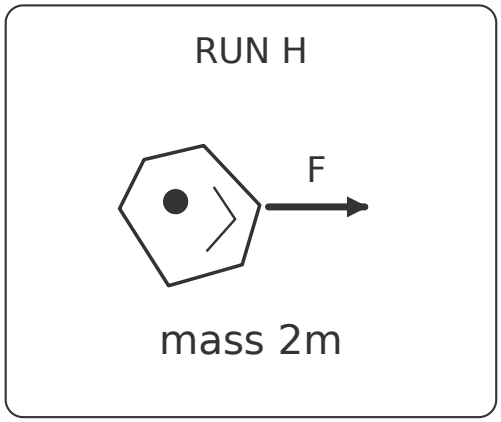
LAB CORRESPONDENT ☐

Explain why equal pushes can give different results. Have your partner check what stayed the same in both tests.

Point to a clue. Keep your first marks; add the update using the color key.

Two asteroid models get the same net force for the same time. Model H has twice the mass of L.





01 OUR FIRST CALL

Which body changes velocity more? Explain your first call from the test card.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

What changed in the test? Give the crew a new comparison.

Switch speakers. Which things stayed the same in the two tests?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks